56% increase in AI-driven attacks. Read findings

Artificial intelligence Statements and reactions

The trends that will shape A

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From quantum to efficiency: The new compute frontier Beyond moc



Anabelle Nicoud
Staff Writer
IBM

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A year in tech can feel like a decade anywhere else.

Think about it: a year ago, we were discussing how ChatGPT wasn't abl
Chinese frontier labs (like DeepSeek-R1) hadn't taken the world by stor

Claude's dedicated coding agent didn't exist yet. IBM's Granite 3.0 had
had just gained traction in the spring, with a notable endorsement from

Meanwhile, in the world of infrastructure, chips and compute resources
advantage.

Over the last few weeks, *IBM Think* spoke with a dozen experts in tech-
insights on what to expect in the year ahead. Each one shared a commc
2026.

“It’s such a crazy time,” Peter Staar, a Principal Research Staff Member
“And it’s only accelerating.”

New agentic capabilities will give way to new possibilities for businesse
la Rick Rubin style with AI creation,” IBM’s Distinguished Engineer Chri
become AI composers, whether you’re a marketer, programmer or PM.’

Many believe efficiency will be the new frontier. “GPUs will remain king
even quantum-assisted optimizers will mature,” Kaoutar El Maghraoui,
of Experts. “Maybe a new class of chips for agentic workloads will emer

After much skepticism around AI’s ROI, AI capabilities will pave new w
models and agents will keep pushing boundaries to conquer enterprise

At the same time, trust and security will become key priorities as many

That’s just the opening act for what’s to come in enterprise tech in the c

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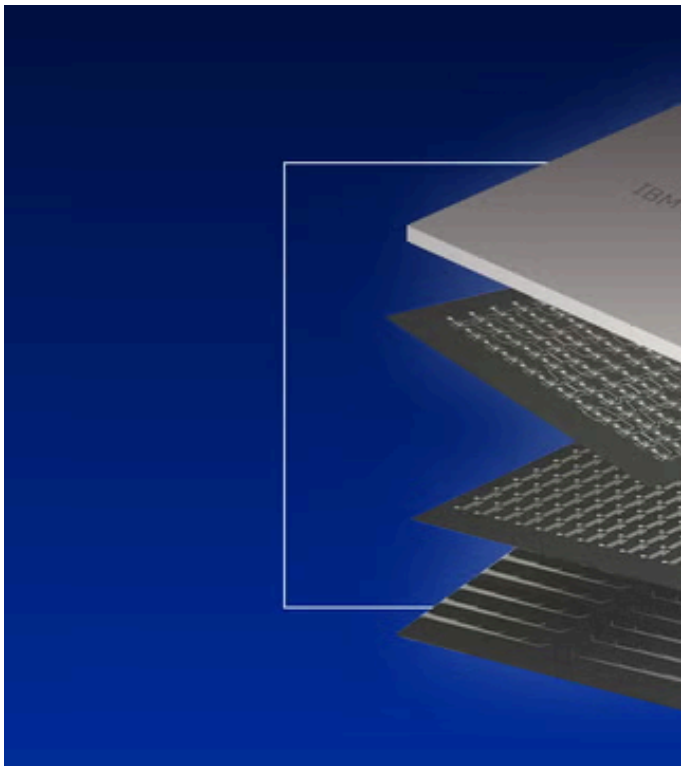
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From quantum to efficiency: The new



Quantum will outperform classical
Strategic Growth & Quantum Partner

IBM has publicly stated that 2026 will mark the first time a quantum co which a quantum computer can solve a problem better than all classica

According to IBM, this milestone will unlock breakthroughs in drug dev industries facing incredibly complex challenges.

"We've moved past theory," Jamie Garcia, Director, Strategic Growth ar the industry's best-available quantum computers for real use cases. WI expect value to increase as quantum continues maturing. And we are s materials discovery and optimization for finance and logistics."

Garcia also highlights the convergence with AI: tools like Qiskit Code As automatically. IBM is building a quantum-centric supercomputing archi performance computing and AI infrastructure, supported by CPUs, GPU

To push this goal into the future, AMD and IBM are exploring how to int efficiently accelerate a new class of emerging algorithms, which are out

Read more on quantum advantage →

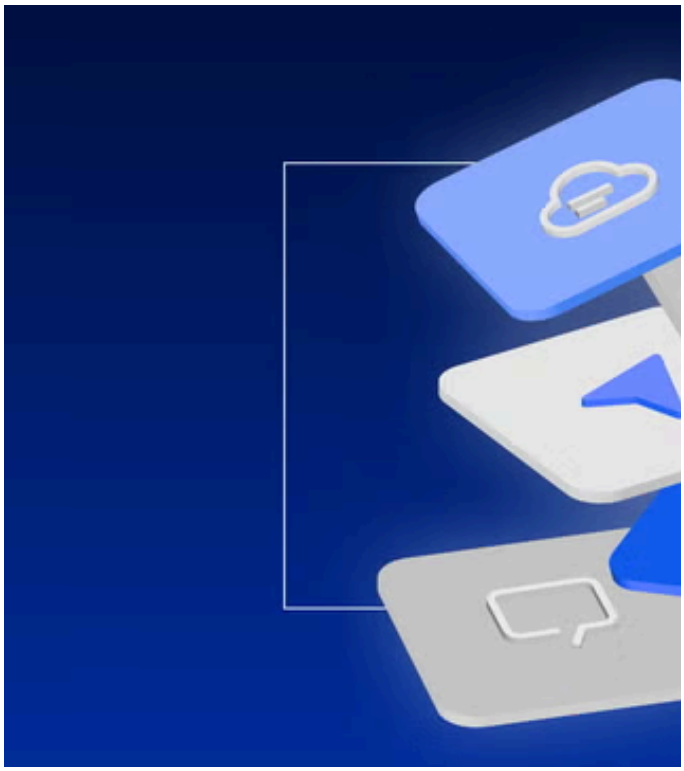
Hardware efficiency will become th
Maghraoui, Principal Research Scien
Enablement, AI Hardware Center, IB

"2026 will be the year of frontier versus efficient model classes," Kaout recent episode of *Mixture of Experts*. Next to huge models with billions accelerators will appear. "We can't keep scaling compute, so the indust

In 2025, demand outran the supply chain, forcing companies to optimiz strategies: scale-up with superchips like H200, B200, GB200—or scale

This will also mean that edge AI will move from hype to reality. And the
king, but ASIC-based accelerators, chiplet designs, analog inference an
“Maybe a new class of chips for agentic workloads will emerge.”

Beyond models: The rise of AI syster



Systems, not models, will define AI
Architect, AI Open Innovation, IBM

In 2026, the competition won’t be on the AI models, but on the system

“We’re going to hit a bit of a commodity point,” Gabe Goodhart, Chief A
Think. “It’s a buyer’s market. You can pick the model that fits your use
be the main differentiator.”

What matters now is orchestration: combining models, tools and workf
explained. “You are talking to a software system that includes tools for
programmatic tasks, and most likely an agentic loop.”

“In 2026, I think we’ll see more sort of cooperative model routing,” Go
delegate to the bigger model when needed. Whoever nails that system-

Gabe Godhart discusses the wins, fails and future of AI code generatio

Agentic parsing will replace monoli
Raymond, Founder & CEO, Unstructu

In 2026, document processing will stop being a one-model job. Instead
pipelines break documents into their parts (titles, paragraphs, tables, ir

“This allows us to reduce computational cost while improving fidelity b
understands it best,” Brian Raymond, Founder and CEO of Unstructured

"The result is a flexible reconstruction layer that synthesizes a precise r
guarantees about structure, lineage and meaning," Raymond said. Unst
Research's Docling in order to accomplish this objective, increasing ove

Next comes agentic parsing. Think of it as a team of domain experts—o
deep semantic profiles and indexing everything across a multidimensio
structure, content and metadata simultaneously and makes previously

Together, these advances point to self-aware enterprise data systems,

See how Unstructured and IBM are helping enterprises get their data A

Cross-functional, cross-channel "su
Hay, Distinguished Engineer, IBM

"We've moved past the era of single-purpose agents," Chris Hay, Distin
episode. In 2024, agents were small and specialized: the email writer, t
plan, call tools and complete complex tasks.

"We're seeing the rise of what I call the 'super agent,'" Hay said.

"In 2026, I see agent control planes and multi-agent dashboards beco
operate across environments—your browser, your editor, your inbox—w
the static software in the user experience and user interface. Expect int
every user an AI composer.

"Whoever owns that front door to the super agent will shape the marke

AI agents will shift from personal as
as everyday users become the new
Strategy Officer, Writer

2026 will be defined by three trends that move AI beyond personal pro
enterprise AI platform for agentic work.

"First, AI is shifting from individual usage to team and workflow orches
workflows, connecting data across departments and moving projects fr

Second, as reasoning capabilities improve, systems won't just follow in
a passive assistant into an active collaborator capable of meaningful pr

Finally, Chung sees the most exciting shift: the democratization of AI ag
moving beyond developers into the hands of everyday business users,"
see a wave of innovation driven by people closest to real problems."

AI will move from tool to teammate
Faro, VP Quantum and AI, IBM Resea

Agentic systems turned LLMs and coding assistants into something mo
Faro, VP of Quantum and AI at IBM Research. He sees software moving
goals and validate progress while autonomous agents execute tasks an

"Software practice will evolve from vibe coding to Objective-Validation
going to define goals and validate while collections of agents autonomo
human approval at critical checkpoints."

This shift will enable the emergence of agentic runtimes to run comple

This will be the foundation for an “Agentic Operating System (AOS),” Fa
and resource governance across agent swarms.

“With disciplined attention to security, resource management, compliar
system agents to reclaim leadership in mission-critical computing,” he

Multimodal AI will interpret the wo
IBM Fellow and Master Inventor, IBM

Generative models need to be multisensory so they can interpret the w
Baughman, IBM Fellow and Master Inventor, in a recent episode of *Mixt*

Baughman has worked with multimodal AI in sports and leads some of
notably. For him, multimodal AI is a trend he expects to see more of in

“These models will be able to perceive and act in a world much more lil
together,” he said. “In the near future, we’re to start seeing these multi
tasks to interpret things, maybe even like complex healthcare cases.”

But autonomy won’t mean removing human oversight. “It’s also import
“so that the human can fine tune and change the skill.”

Agent-to-agent communication will
Blair, Director of Incubation and Tech

It’s been just a year since Anthropic launched MCP, alongside IBM’s AC
be the year where all multi-agent systems move into production, IBM’s
protocol maturity and convergence.

“2026 is when these patterns are going to come out of the lab and into
Both projects were contributed to the Linux Foundation.

The Linux Foundation recently announced the formation of the Agentic
that MCP has come under open governance,” Blair said. “Openly goverr
more innovation and more solutions.”

The A2A project is about to hit its first major release. “We’re already se
card to describe an entity, whether it’s a tool or resource in MCP or an a

Blair sees this unified card as a catalyst for interoperability and the opp
agentic systems.

“I’m excited to get to the next level where we’re really speaking about v
agents.”

Read more on Agent Stack →

Enterprise AI, reinvented



AI will deliver real ROI ... securely | CEO, Atolio

Atolio provides a secure and private AI platform for enterprises. The ch
experiment with new tech while also mitigating the risk of losing contro

“The most significant trend we see emerging next year is the shift from
deployments with real ROI expectations within enterprises,” David Lan

“Data leaks continue to erode enterprise trust,” he said. “The unsolved
makes data sovereignty and first-class permissioning non-negotiable re

The solution isn’t bigger models, but smarter data. As Lanstein puts it: ‘
aware structured data to generate intelligent, relevant and trustworthy

“What excites me most is the convergence necessary to make this happ
advancements and solutions that understand context and user needs n

AI agent adoption will force compa access management strategy | Shlo AuthMind

Founded in 2020, AuthMind is tackling one of the toughest problems in
identity’s access and activity so they can stop attacks before they start.

“In the coming years, agentic AI and other non-human identities will ou
CEO and Cofounder at AuthMind, told *IBM Think*.

The shift will redefine enterprise security and governance. “This is now
acting the way it was intended to, increasing both productivity and secu
no longer just deploying models; it’s managing identity with new users:

For enterprises, getting the advantage in this context means answering

Discovering, observing and protecting not just every human but also ev
adoption. “I am super excited to follow the companies that master this
said.

Machine automation will become tr
CEO, Rohirrim

“The most powerful trend I see for next year is AI tackling complex ent
native startup focused on complete procurement ecosystems, in an int
system that can execute deep tasks, end to end.”

Generative and agentic systems will interpret intent, search across vast
achieved. “That shift creates entirely new categories of platforms, and
human or a single application can hold in mind,” he explained. “It is tru

Transformers made this possible. “They gave us systems that can abso
nuance and precision with effective guardrails in place,” Aberle said. “W
to directly influencing outcomes.”

In procurement, that means tracking requirements, spotting gaps early
faster decisions.

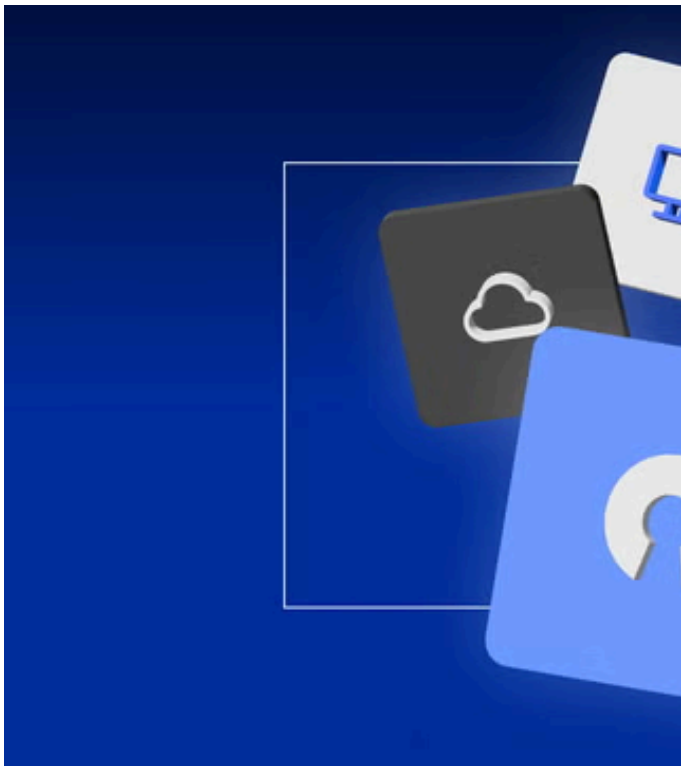
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Open source shapes the future



Open source will continue its divers
Matt White, Executive Director, PyTo

A year ago, Matt White, Executive Director of the PyTorch Foundation, p

“The industry validated the thesis that smaller, domain-optimized mod
in distillation, quantization and memory-efficient runtimes pushed infe
and data-sovereignty needs.”

According to White, three forces will be defining open-source AI in 202
reasoning-tuned releases; interoperability as a competitive axis, as frar
governance, with security-audited releases and transparent data pipeli

“As agentic systems emerge, PyTorch’s role as a common substrate for
“Developers need flexible tooling for multimodal reasoning, memory cc
source thrives.”

Scaling will hit its limits, and physic
Staar, Principal Research Staff Memb
the Technical Steering Committee of
AI and Data

IBM’s Peter Staar predicts 2026 will mark a shift in AI research prioritie
going to pick up,” he said. While large language models remain dominar
scaling. “People are getting tired of scaling and are looking for new idea

Staar sees a lot of interest for AI that can sense, act and learn in real er
the next frontier for innovation.

At the same time, Staar believes that open-source AI will continue to sl
closed, and the ones catching up go open,” he said.

AI will shift from giant models to dc Anthony Annunziata, Director of Ope Alliance

2024 ended on a high note for open-source AI with Meta's Llama mode grown a lot, with smaller, domain-specific models achieving impressive DeepSeek's models. Anthony Annunziata, Director of Open Source AI a

"We're going to see smaller reasoning models that are multimodal and *IBM Think*.

Advances in fine-tuning and reinforcement learning also mean that ente and efficient models. "Instead of one giant model for everything, you'll more so—when tuned for the right use case," he said.

Open source, agentic AI will accelerate this trend. "General-purpose ag said. "You need domain-enriched models and architectures that reflect

Open-source AI is a necessity. "If you believe we're heading toward an standards of interaction must be open, he said." "Otherwise, you end u

Join the AI Alliance

Trust as a strategy



Decentralized AI will move beyond prototypes | Tomás Hernando Kofman, Cofounder, Not Diamond

"The massive middle of the enterprise bell curve begins to move from experimentation to production-grade systems," said Tomás Hernando Kofman, Cofounder of Not Diamond, a multi-model AI infrastructure platform, in an interview with *IBM Think*.

That transition won't be easy: "AI teams will have to invest heavily in evaluation, reliability, optimization, efficiency, scalability and

This will take serious coordination and resources. If companies don't make the investment, they'll end up stuck: not having the right capabilities means the systems aren't useful, and that lack of utility just reinforces the problem.

On the frontier, the challenges look different. "I believe the field will grapple with three major hurdles: continuous learning, memory and scalability," Kofman said.

Work will happen at both the model architecture level and in agentic systems.

"We'll begin to see decentralized networks of agents that can learn from each other, share information and retain important knowledge over long horizons—weeks, months, even years," he said. "These systems will improve dynamism and continuous improvement while allowing agents and models to specialize into efficient, focused capabilities."

Collaborative defenses will fight weaponized AI | Ben Colman, CEO & Cofounder, Reality Defender

No single entity can solve the deepfake and weaponized AI crisis, especially as new threat vectors—like weaponized AI agents—emerged at the tail end of this year, said Ben Colman, CEO & Cofounder of Reality Defender, a cybersecurity company offering deepfake detection tools. The rapid evolution of generative AI demands a collaborative ecosystem.

"Strategic partnerships are essential, not just for bolstering defense, but for anticipating the next wave of sophisticated models and industry-specific vulnerabilities," he told *IBM Think*.

"I see these partnerships in our industry between ourselves, others working on different aspects of a similar problem and so on—happening as fast as advancements in AI, if not faster."

Colman observes a shift toward a layered security model. "By stacking different defenses, gaps in one layer are covered by another to create an impenetrable shield," he said.

Integration will define the next phase. "When these emerging technologies couple with detection platforms like ours, the result is a comprehensive 'defense in depth' strategy," he said. "This ensures organizations are protected across all media formats and entry points, for all use cases, and in all toolsets—rather than relying on a single point of failure."

AI resilience will become critical | Anthony Marshall, Senior Director and Vice President, IBM Institute for Business Value

Organizations can't afford to have their AI projects interrupted—but there's only so much that business leaders can control. AI sovereignty—the ability to govern AI systems, data and infrastructure without relying on external entities—has become mission-critical, said Anthony Marshall, Senior Director and Vice President, IBM Institute for Business Value (IBV).

For 93% of executives surveyed by the IBV, factoring AI sovereignty into business strategy will be a must in 2026. "This isn't just box-checking," said Marshall.

Half of executives worry about over-dependence on compute resources in certain regions (a concern especially high among business leaders in the Middle East and APAC), and many believe that leaning on those resources can introduce a wide range of risks. Think data breaches, loss of access to data and intellectual property theft.

Transparency and trust will also remain priorities. "Both regulators and consumers ask organizations to explain how AI agents come to specific decisions. Organizations must design agents that can show their work, for even the most complex outputs," said Marshall.

That means building sovereignty through modularity—architecting AI environments so workloads, data and agents can shift among trusted regions and providers.

"Continuous monitoring is essential to detect and address model drift before it compromises performance or introduces bias," said Marshall.

Discover the 5 business trends for 2026 in IBV's latest report →

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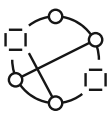
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